

```
# empty list
```

```
my_list = []
```

```
# list of integers
```

```
my_list = [1, 2, 3]
```

```
# list with mixed data types
```

```
my_list = [1, "Hello", 3.4]
```

```
#multiple datatype assign
```

```
n_list=[2.9,3000,'Hello',"Welcome to python list ", True]
```

```
print(n_list)
```

```
# Nested List Example1
```

```
n_list = ["Happy", [2, 0, 1, 5]]
```

```
# Nested indexing
```

```
print(n_list[0][3])
```

```
print(n_list[1][2])
```

```
# Nested List Example2
```

```
my_list = ["mouse", [8, 4, 6], ['a','b','c']]
```

```
print(my_list)
```

```
print(my_list[0])
```

```
# Nested indexing
```

```
print(my_list[1][1])
```

```
print(my_list[2][2])
```

```
# Negative indexing in lists
```

```
my_list = ['p','r','o','b','e']
```

```
print(my_list[-1])
```

```
print(my_list[-5])
```

Appending and Extending lists in Python

```
odd = [1, 3, 5]
odd.append(7)
print(odd)
```

```
odd.extend([9, 11, 13])
print(odd)
```

Concatenating and repeating lists

```
odd = [1, 3, 5]

print(odd + [9, 7, 5])
```

```
print(["re"] * 3)
```

```
thislist = ["apple", "banana", "cherry"]
print(len(thislist))
```

```
thislist = ["apple", "banana", "cherry"]
thislist.remove("banana")
print(thislist)
```

```
fruits = ["apple", "banana", "cherry", "apple"]
```

```
x = fruits.count("apple")
print(x)
```

```
fruits = [1, 4, 2, 9, 7, 8, 9, 3, 1]
x = fruits.count(9)
```

```
print(x)
```

```
# Program to demonstrate
```

```
lst = [10, 20, 30, 40]
```

```
# Create a copy of the list
```

```
new_lst = lst.copy()
```

```
print("Original List :", lst)
```

```
print("Copied List  :", new_lst)
```

```
lst = [10, 20, 30, 40, 50]
```

```
value = 30
```

```
position = lst.index(value)
```

```
print("List  :", lst)
```

```
print("Value  :", value)
```

```
print("Index  :", position)
```

```
# tuple of strings
```

```
my_data = ("hi", "hello", "bye")
```

```
print(my_data)
```

```
# tuple of int, float, string
```

```
my_data2 = (1, 2.8, "Hello World")
```

```
print(my_data2)
```

```
# tuples inside another tuple
```

```
# nested tuple
```

```
my_data4 = ((2, 3, 4), (1, 2, "hi"))  
print(my_data4)
```

output

```
[2.9, 3000, 'Hello', 'Welcome to python list ', True]
```

```
p
```

```
1
```

```
['mouse', [8, 4, 6], ['a', 'b', 'c']]
```

```
mouse
```

```
4
```

```
c
```

```
e
```

```
p
```

```
[1, 3, 5, 7]
```

```
[1, 3, 5, 7, 9, 11, 13]
```

```
[1, 3, 5, 9, 7, 5]
```

```
['re', 're', 're']
```

```
3
```

```
['apple', 'cherry']
```

```
2
```

```
2
```

```
Original List : [10, 20, 30, 40]
```

```
Copied List  : [10, 20, 30, 40]
```

```
List   : [10, 20, 30, 40, 50]
```

```
Value  : 30
```

```
Index  : 2
```

```
('hi', 'hello', 'bye')
```

```
(1, 2.8, 'Hello World')
```

((2, 3, 4), (1, 2, 'hi'))